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Exploring long- and short-term knowledge state graph representations with adaptive fusion for knowledge tracing

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ABSTRACT

Knowledge Tracing (KT) is an important research area in online education that focuses on predicting future academic performance based on students' historical exercise records. The key to solving the KT problem lies in assessing students' knowledge states through their responses to concept-related exercises. However, analyzing exercise records from a single perspective does not provide a comprehensive model of student knowledge. The truth is that students' knowledge states often exhibit long- and short-term phenomena, corresponding to long-term knowledge systems and short-term real-time learning, both of which are closely related to learning quality and preferences. Existing studies have often neglected the learning preferences implied by long-term knowledge states and their impact on student performance. Therefore, we introduce a hybrid knowledge tracing model that utilizes both long- and short-term knowledge state representations (L-SKSKT). It enhances KT by fusing these two types of knowledge state representations and measuring their impact on learning quality. L-SKSKT includes a graph construction method designed to model students' long- and short-term knowledge states. In addition, L-SKSKT incorporates a knowledge state graph embedding model that can effectively capture long- and short-term dependencies, generating corresponding knowledge state representations. Furthermore, we propose a fusion mechanism to integrate these representations and trace their impact on learning outcomes. Extensive empirical results on four benchmark datasets show that our approach achieves the best performance for KT, and beats various strong baselines with a large margin.

1. Introduction

Enabled by various information technologies, online learning platforms have evolved significantly over time (Domingue et al., 2014; Vardi, 2012). Learning platforms enhance education by providing a richer, more flexible environment that complements traditional teaching methods. These platforms offer significant advantages, such as flexibility, convenience, rich resources, and diverse curriculum options, making them highly attractive in modern education (Cole et al., 2017; Liu, Huang et al., 2023; Zhang et al., 2025). Meanwhile, the capability of platforms to automatically monitor students' learning progress and knowledge

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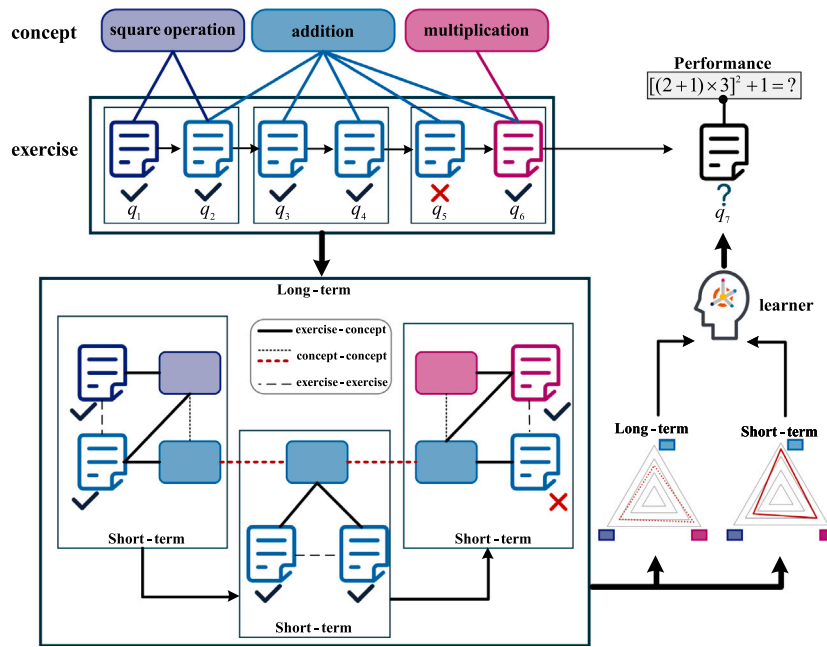


Fig. 1. A simple schematic of KT. Based on a student's learning sequence of concept-related exercises, KT predicts responses to the final exercise and indirectly models the student's performance of knowledge-concept mastery over long- and short-term knowledge states.

state, dynamically adjust teaching progress, and optimize a personalized learning experience tailored to students' needs becomes crucial. One of the key research areas to help address these issues is knowledge tracing (KT). Typically, KT attempts to estimate and track a student's knowledge state for mastering concepts based on historical learning records and then predicts a student's future academic performance based on the estimated knowledge state. Taking the example shown in Fig. 1, to predict a student's academic performance during the response exercise q_7 , the KT system evaluates the knowledge states (e.g., mastery of knowledge concepts, such as *squaring*, *addition*, *multiplication*) based on the student's historical exercise records from the response exercises q_1 to q_6 (Anderson et al., 1990; Shen et al., 2024).

KT can help teachers to dynamically adjust teaching schedules, learning materials and provide students with learning services that support personalized and adaptive learning in online learning systems (Wang, Wu et al., 2019). Since KT tasks are too abstract and difficult to define accurately, researchers often convert it into prediction tasks based on students' knowledge state representations. In the literature, previous KT models have evolved from traditional approaches such as probabilistic models (Baker et al., 2008) to deep learning-based approaches (Gan et al., 2020; Ghosh et al., 2020). Deep learning-based approaches are based on deep neural networks such as deep knowledge tracing (DKT) using recurrent neural networks (RNNs) (Piech et al., 2015), dynamic key-value memory networks (DKVMN) using memory networks (MNs) (Zhang, Shi et al., 2017) and graph knowledge tracing (GKT) using graph neural networks (GNNs) (Nakagawa et al., 2019). Among them, GNN-based KT exhibits superior performance because they model concepts or exercises as nodes in a unified graph representation learning framework, which effectively simulates the structural relationships between different concepts and the interactions between different exercises to model students' knowledge states.

Although these existing approaches have made great advancements, they are still unsatisfactory. The main limitation of existing research is that it focuses on constructing students' knowledge state embedding based on the knowledge concepts and related exercises they have learned in a short period of time, analyzes the short-term iterative process, ignoring the comprehensive knowledge system constructed by the students in the long-term learning process, and lacking a more holistic analysis of the knowledge states. Meanwhile, existing research on knowledge state modeling is relatively limited, which restricts its applicability and flexibility in handling complex knowledge state embedding. In summary, the potential for comprehensive modeling of knowledge states to improve KT performance remains to be explored.

In practice, we observe in conjunction with cognitive load theory (Krathwohl, 2002; Paas et al., 2016) that students generally possess relatively stable long-term knowledge states and dynamic short-term knowledge states. Long-term knowledge states reflect students' deep understanding of knowledge and overall learning preferences, corresponding to the stable knowledge system. In contrast, the short-term knowledge state reflects recent learning preferences and outcomes, usually shows the real-time change phenomenon. This implies that accurately modeling a student's knowledge state requires considering both long- and short-term perspectives. The modeling approach should not only comprehensively assess the student's overall learning process to capture their long-term knowledge state, but also focus on their recent learning activities and performance to update their short-term knowledge state in real-time. This method is crucial for improving the accuracy of KT prediction.

As shown in Fig. 1, our goal is to predict whether the student can correctly response to an exercise q_7 involving three knowledge concepts. Based on the student's exercise records, we construct graphs covering concepts and exercises in terms of long- and short-term knowledge states. From the **long-term perspective**, the student's overall mastery of the concept of *squaring* remains good, despite the long time gap that has elapsed since their last exposure to the concept. This suggests that the student has not forgotten the concept of *squaring* calculations, but has further reinforced and consolidated this knowledge point through the complex interplay between subsequent exercises and related knowledge concepts. However, from the **short-term perspective**, the student's recent learning preferences focus on the concepts of *addition* and *multiplication*, with less exercise in the concept of *squaring*. From this single short-term perspective, it appears that the student lacks a sufficient knowledge base to answer the final question correctly. This suggests that relying solely on iterative updates of short-term knowledge states to track student performance is insufficient.

Actually, jointly modeling knowledge states from both long-term and short-term perspectives is a challenging task. **First**, most existing methods fail to explore long-term knowledge states explicitly and lack explicit modeling of their long-term dependencies. They primarily rely on recurrent sequence models to capture the short-term interactions and iterative processes of students' knowledge states, but neglect the influence of long-term historical sequences, leading to suboptimal performance. It is urgent to model the long-term dependencies of knowledge concepts and their corresponding exercises that occur at the long time ago. **Second**, to make accurate predictions of student's future performance, long- and short-term states of knowledge should be considered, and their importance should be assessed based on the individual student. When students continuously learn similar knowledge concepts, short-term knowledge states become more significant. However, when the focus shifts to different knowledge concepts, students' knowledge states are primarily driven by long-term knowledge states. Thus, it is crucial yet challenging to adaptively integrate these two aspects based on the specific context of the knowledge concepts that students encounter in order to predict future performance.

To address these challenges, we propose a hybrid model for knowledge tracing based on students' long- and short-term knowledge states (L-SKSKT), aiming to explore the association between these knowledge states and predicted performance. **First of all**, we model the knowledge states using the graph where nodes represent the corresponding concepts and exercises, while the edges denote the complex relationships among them. Within the graph, the short-term learning perspective is modeled as a series of subgraphs based on the explicit relationships between concepts and exercises. On the contrary, the long-term learning perspective, modeled by unifying all short-term subgraphs, summarizes the students' whole learning experiences based on the exercises and concepts across all learning states. Unlike each short-term subgraph, the long-term unified graph introduces the more complex relationships (e.g., *concept-exercise*, *concept-concept*), which can better simulate the knowledge states throughout the whole learning process. **Then**, we design a knowledge state graph embedding model, and unlike previous studies that have modeled long- and short-term knowledge states in a unified framework, we construct their corresponding knowledge state representations hierarchically and independently. Based on multiple relationships in a heterogeneous graph, we can ensure the validity of long-term dependency features between nodes and construct more iconic representations. **Finally**, we develop an adaptive fusion module that adjusts the weights of long- and short-term knowledge states according to the specific learning scenarios that students face during prediction, and fuses the knowledge state representations to improve the accuracy of performance prediction. In summary, our contributions are as follows:

- We design a knowledge state graph construction method, which can transform the original exercise records into long- and short-term knowledge state graphs through the complex multiple relationships among the exercise concepts.
- We propose a knowledge state graph embedding model to construct corresponding knowledge state representations hierarchically and independently. Meanwhile, we also design an adaptive fusion module to fuse long- and short-term representations adaptively.
- We conduct extensive experiments on four KT benchmark datasets. The experimental results show that our proposed approach outperforms strong baselines under different evaluation metrics.

The structure of this paper is outlined as follows. Section 2 discusses the relevant literature pertaining to this study. In Section 3, we provide a formal definition of the research task along with the corresponding notations. Section 4 introduces the proposed L-SKSKT model. The details of the experimental results are presented in Section 5. Finally, we summarize our findings in Section 6.

2. Related work

In this section, we review existing KT research methods and graphical representation learning theory, exploring their applications in educational settings and offering an analysis of their advantages and limitations.

2.1. Knowledge tracing

Effective tracing of students' learning progress is a key challenge in the educational context, which requires interactions between students and instructional materials, as well as recording of exercises generated during the period. KT plays an important role in solving this challenge and shows its great potential in realizing intelligent education. In general, previous studies on KT can be divided into four categories (Abdelrahman et al., 2023): traditional, seq-based, attention-based, and graph-based models. Table 1 shows the summary of different KT models.

The classic traditional KT model is represented by Bayesian knowledge tracing (BKT) (Corbett, 2000), which utilizes hidden Markov models (Baker et al., 2008) and Bayesian belief networks (Villano, 1992) to achieve its KT task. Meanwhile, BKT has several belief theory variants (Käser et al., 2017; Yudelson et al., 2013). Item response theory (IRT) (Rasch, 1981) is another traditional offshoot of KT, and the knowledge tracing machine (KTM) proposed by Vie (Sweeney et al., 2016) provides a general framework

Table 1
A Summary of the KT models.

Cate	Model	Learning Model
Traditional	BKT (Corbett, 2000)	HMM/BN
	KPT (Huang et al., 2020)	HMM/BN
	IRT (Rasch, 1981)	LR
	KTM (Sweeney et al., 2016)	FM
Seq-based	DKT (Piech et al., 2015)	RNN/LSTM
	AT-DKT (Liu, Liu et al., 2023)	RNN/LSTM
	DKT+(Yeung & Yeung, 2018)	RNN/LSTM
	DKT-DSC (Minn et al., 2018)	RNN/LSTM
	LPKT (Shen et al., 2021)	RNN/LSTM
	LBKT (Xu, Huang et al., 2023)	RNN/LSTM
	FDKT (Liu, Bu et al., 2024)	RNN/LSTM
	DKVMN (Zhang, Shi et al., 2017)	KVMN
Attention-based	SAKT (Pandey & Karypis, 2019)	FFN+MSA
	SAINKT (Choi et al., 2020)	FFN+ED+MSA
	AKT (Ghosh et al., 2020)	FFN+MSA
Graph-based	GKT (Nakagawa et al., 2019)	GNN
	GIKT (Yang et al., 2021)	GNN+RNN
	DGEKT (Cui et al., 2024)	GNN+GRU

for student modeling based on factorization machines (FM). BKT and IRT have a history of integrating each other's research, and it is a common research approach to extend IRT by incorporating Bayes' theorem (Khajah, Huang et al., 2014; Khajah, Wing et al., 2014; Wilson et al., 2016).

In contemporary times, numerous researchers have made significant contributions to the domain of KT. The evolution progress of KT models is from early traditional Bayesian inference-based approaches (Baker et al., 2008) to parametric factor analysis (Cen et al., 2006; Pavlik Jr et al., 2009), and presently, to sophisticated methods grounded in deep learning (Gan et al., 2020; Ghosh et al., 2020). The research trajectory of KT remains vibrant. In a broad sense, the currently popular models attempt to delve into the dynamics of learning in the exercise records. In this context, student feedback and related exercise information are integral components of the sequence and have been extensively explored in the data mining process (Liu et al., 2021; Piech et al., 2015; Zhang, Shi et al., 2017). However, it is insufficient to model exercises and concepts from a single feature. Deep mining of exercise records ignores other relevant features collected by computerized learning platforms that are closely related to the knowledge state of the student (Zhang, Xiong et al., 2017).

The rapid progress in deep learning (LeCun et al., 2015) has prompted many KT studies to adopt deep learning techniques. Currently, KT models supported by deep learning techniques can be categorized into sequence-based models, attention-based models, and graph-based models. Deep Knowledge Tracing (DKT) is the pioneering work that began by integrating KT with deep learning principles (Piech et al., 2015). Researchers have also extended DKT to explore and study many variants (Collier & Beel, 2018; Liu, Liu et al., 2023; Minn et al., 2018; Xiong et al., 2016; Yeung & Yeung, 2018). Like the dynamic key-value memory network (DKVMN) (Zhang, Shi et al., 2017) enhances DKT with dynamic memory matrices for keys and values. Attention mechanisms also play a significant role in KT innovations. Self-attentive knowledge tracing (SAKT) introduces self-attention mechanisms (Pandey & Karypis, 2019) and attentive knowledge tracing (AKT) integrates the Rasch model enhanced problem representation (Ghosh et al., 2020). Meanwhile, separated self-attentive neural knowledge tracing (SAINT) (Choi et al., 2020) builds question-embedded sequences and response-embedded sequences by dividing the students' interaction sequences into question-embedded and response-embedded sequences, respectively, and feeds them into encoders and decoders for further analysis. Pu et al. proposed the enhancing locality for attentive knowledge tracing (ELAKT) model (Pu et al., 2024), which improves the prediction performance of KT by improving the attention mechanism and introducing a prediction correction module. The advent of graph neural networks (GNN) in deep learning has introduced novel possibilities in KT by modeling students' data as graphs. Graph-based Knowledge Tracing (GKT), proposed by Nakagawa et al. (2019). GKT reformulates the KT problem as a time series node-level classification problem, employing standard graph learning techniques like message-passing GNNs (Scarselli et al., 2008). Following GKT, various GNN-based KT models have emerged, combining them with features such as interaction relations (Yang et al., 2021) and structural information (Tong et al., 2020), resulting in the development of exemplary models. Recently, research workers have also proposed a new approach for knowledge tracing called the dual graph integrated learning method (DGEKT) (Cui et al., 2024), which captures heterogeneous motion-concept associations and interaction transitions by building a dual graph structure of student learning interactions. The online knowledge distillation technique is introduced to improve the performance of KT significantly.

Currently, researchers have analyzed exercise records from multiple perspectives (like combining behaviors (Xu, Huang et al., 2023; Zhang, Shi et al., 2017), environments (Shen et al., 2021; Zhang, Xiong et al., 2017; Zhao et al., 2023), etc.) to enhance the performance of KT models. For example, Minn et al. (2022) proposed Interpretable Knowledge Tracking (IKT) to predict students' future performance based on latent traits. Schultz and Arroyo (2014) pointed out that students' performance depends on knowledge and motivation. Cui, Chen et al. (2023) proposed a multi-relational transformer knowledge tracing (MRT-KT) to model fine-grained topic-response interactions through relational coding. Wang and Heffernan (2012) analyzed initial reaction time to predict student performance. The knowledge proficiency tracing (KPT) model incorporates educational prior knowledge,

associates exercises with knowledge vectors and captures dynamic changes in proficiency using learning curves and forgetting curves (Huang et al., 2020). Other models such as the learning process-consistent knowledge tracing model (LPKT) (Shen et al., 2021), which incorporates response time, and the sequence-flexible knowledge tracing (SFKT) model (Zhang, Zhu et al., 2023), which introduces data enhancement and comparison learning tasks. In addition, some studies focus on the impact of behavioral data on KT. Mongkhonvanit et al. (2019) incorporated video viewing behaviors (e.g., playback speed and fast-forwarding) into deep knowledge tracing (DKT) for MOOCs. The learning behavior-oriented knowledge tracing (LBKT) (Xu, Huang et al., 2023) systematically analyzes the effects of learning speed, attempts, and cues on knowledge states. Another study proposed fuzzy deep knowledge tracing (FDKT) (Liu, Bu et al., 2024), which formalizes continuous scores into fuzzy scores by means of a fuzzification module and uses a fuzzy inference module to derive students' cognitive abilities, thus predicting future grades.

It must be acknowledged that sequence-based and attention-based models have been successful, but they tend to ignore the long-term learning process of the student, and there is singularity in the modeling of exercise records. Also, about the GNN-based models, focus on conceptual modeling, ignoring the importance of closely related exercise. Meanwhile, although there have been works that are aware of modeling from multiple perspectives, KT models for modeling long- and short-term knowledge states are still mostly based on sequence modeling, which fails to capture the complex multiple associations between exercises and concepts, as well as temporal correlations. Current research lacks a comprehensive approach to characterize the complexity of students' knowledge acquisition and problem-solving abilities over time. In contrast to existing research, the model in this paper employs graph-structured modeling of exercise records to capture the long- and short-term dependency characteristics of exercise concepts at different moments, and to explore the weights of specific knowledge states that play a key role in providing students with cognitive support for themselves when answering questions.

2.2. Graphical representation learning

GNNs are developing rapidly and gradually becoming a powerful tool for processing unstructured data, which are applied in social network analysis, bioinformatics and other fields. In the modeling of node complex relationships, GNN also reflects its strong compatibility and scalability. Graph convolutional neural network (GCN) (Kipf & Welling, 2016) and graph attention neural network (GAT) (Veličković et al., 2017) are two major branches of GNN, the former realizes the updating of node features through convolution operation, and the latter uses the attention mechanism to discover the dynamic modeling of the relationship between nodes. Also, there have been recent prospective studies on hypergraph graph neural networks (HGNN) (Feng et al., 2019) as well as directed graph neural networks (DGNN) (Shi et al., 2019).

Dai et al. (2024) combined anatomical patterns in segmentation methods in the medical image domain to enhance the representation of nodes in the knowledge graph by using the idea of mutual information maximization. Meanwhile, the introduction of temporal knowledge graph (TKG) enables GNN to model temporal feature data, and in recent years, the emergence of models combining historical interaction-dependent attributes and potential influence associations (Xu, Ou et al., 2023) and models based on temporal perception and local-global connectivity (Liu, Wen et al., 2024) enhances the node representation and graph representation capabilities of the model. The GNN-based research shows great advantages in long- and short-term modeling. For example, Liu, Chen et al. (2023) proposed a GNN-based long and short-term preference model (GLSP), which captures users' long- and short-term preferences in time and space. Zhang, Xia et al. (2023) based on the basic concept of GNN, proposed a hierarchical relational graph neural network to learn the long- and short-term representations for TKG reasoning (HGLS), which captures the long- and short-term representations of global and local information in TKG.

At the same time, the graph modeling analysis based on MOOCs has gained great advantages (Deng et al., 2023; Ma et al., 2024; Wang et al., 2022). Graph learning has similarly advanced KT research. Based on graph representation learning, generating learning exercises and concept embedding to enhance DKT respectively, assisting the task to predict associations between exercises and knowledge concepts is an innovative initiative (Wang, Ma et al., 2019). Tong et al. (2020) utilized the prerequisites and similarity relations in the conceptual knowledge structure, while considering the temporal dynamics of learning sequences and the spatial organization of the knowledge structure, and proposed a structure-based KT framework.

Despite impressive progress, sequence structure-based KT studies are unable to intuitively model concept-exercise associations within knowledge states. Meanwhile, graph-based approaches for exercise and concept embedding have typically learned single-relationship embedding, which may not adequately capture the complexity of knowledge states when combined into representations of learning interactions. In contrast to the previous works (Nakagawa et al., 2019) that encode the graph structures with a single type of relational information, our approach utilizes the graph structures to capture the multifaceted nature of exercise-concept associations, fully capturing the dynamic structure and evolution of students' knowledge states.

2.3. Difference to existing methods

Unlike all the approaches described above, our proposed L-SKSKT simultaneously considers students' long- and short-term knowledge states, captures the structural information features in the knowledge states by graph modeling, and predicts learners' future performance through adaptive fusion.

We note that previous studies (Chen et al., 2023; Xu et al., 2022) also considered students' long- and short-term knowledge states. Compared to Chen et al. (2023) and Xu et al. (2022), our model is significantly different in two aspects: (1) **Capturing the structural information that exists between exercise concepts in students' exercise records:** Chen et al. (2023) and Xu et al. (2022) rely on sequential modeling to represent students' knowledge states, which leads to a significant loss of structural

Table 2
Notations and descriptions.

Notation	Descriptions
$\mathcal{X}_{\text{Short}}$	Student's short-term knowledge states
$\mathcal{X}_{\text{Long}}$	Student's long-term knowledge states
m	The short-term window
M	The long-term window
$\mathcal{G}_s$	Short-term knowledge states graph
$\mathcal{G}_l$	Long-term knowledge states graph
$\mathbf{h}_{i,l}, \mathbf{h}_{i,s}$	Node embedding in $\mathcal{G}_l$ and $\mathcal{G}_s$
$\mathbf{h}_{\mathcal{G}_s}$	$\mathcal{G}_s$'s embedding
$\mathbf{h}_{\text{Short}}^{t+1}$	Short-term knowledge state representation
$\mathbf{h}_{\text{Long}}^{t+1}$	Long-term knowledge state representation
ρ	Combined weights in the adaptive fusion
$\mathbf{h}^{t+1}$	The unified representation
y'	The predicted response for time $t+1$

information inherent in these states. In contrast, our approach introduces a knowledge state graph construction method that captures the structural relationships among exercise concepts over time. By modeling these complex relationships, our method effectively addresses the structural information loss observed in previous works. (2) **The capability of capturing short-term knowledge state iterations as well as explicitly modeling long-term dependencies through hierarchical independent modeling of long- and short-term knowledge state representations:** Xu et al. (2022) divides the exercise records into multiple sub-sequences based on timestamps, modeling the last sub-sequence as the short-term knowledge state and the entire record as the long-term knowledge state. However, Xu et al. (2022) has two limitations: (a) Modeling only the last sub-sequence overlooks the iterative nature of short-term knowledge states; (b) the sequential modeling approach restricts the explicit representation of long-term dependencies in knowledge states. Our L-SKSKT model addresses these limitations by employing hierarchical independent graph embedding to assign distinct graph modeling strategies to different knowledge state graphs. Our approach not only captures the iterative nature of short-term knowledge states but also explicitly models the long-term dependencies within long-term knowledge states.

3. Problem definition

In this section, we give a formal definition of the KT task. For convenience, some important notations are listed in Table 2.

Definition 1 (Knowledge State). Let $\mathcal{E}$, $\mathcal{Y}$, and $\mathcal{C}$ represent all exercises learned by the students, as well as the corresponding responses and the knowledge concepts. An exercise record at learning time t is defined as a triple $x_t = (e, y, c)$, indicating a student has correctly answered the exercise e and has mastered the corresponding knowledge concept c if the response $y = 1$, and vice versa. All exercise records at learning time t consist of a knowledge state $\mathcal{X}_t = \{(e, y, c) | e \in \mathcal{E}, y \in \mathcal{Y}, c \in \mathcal{C}\}$. The history learning record $\mathcal{X}$ is defined as a sequence of knowledge states with different learning timestamps, e.g., $\mathcal{X} = \{\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_t\}$. Based on $\mathcal{X}$, We define the short-term historical knowledge state as $\mathcal{X}_{\text{Short}} = \{\mathcal{X}_{t-m}, \mathcal{X}_{t-m+1}, \dots, \mathcal{X}_t\}$, while the long-term history knowledge state is denoted as $\mathcal{X}_{\text{Long}} = \{\mathcal{X}_{t-M}, \mathcal{X}_{t-M+1}, \dots, \mathcal{X}_t\}$, usually $M \gg m$.

Definition 2 (Knowledge Tracing). Given the history learning records $\mathcal{X} = \{\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_t\}$, the task of KT aims to track the students' long- and short-term knowledge states and predict the performance on future exercise records $\mathcal{X}_{t+1}$ at learning time $t+1$:

$$\mathcal{X}_{t+1} = F(\mathcal{X}_{\leq t}) \quad (1)$$

where $F()$ represents the predictive function from $\mathcal{X}_{\leq t}$ to $\mathcal{X}_{t+1}$.

4. Method

The overall framework of our proposed L-SKSKT is illustrated in Fig. 2. L-SKSKT consists of three components: (a) knowledge state graph construction; (b) knowledge state graph representation learning; (c) adaptive fusion for long- and short-term knowledge state representations. Next, we will describe each component in details.

4.1. Knowledge state graph construction

The crucial step for the KT task is how to effectively model the historical exercise records. In the literature, sequence-based models (e.g., DKT, RCKT, etc.), attention-based models (e.g., AKT, ELAKT, etc.), and graph-based models (e.g., GKT, DGEKT, etc.) have been widely investigated. Among them, GKT (Nakagawa et al., 2019) is the first work that constructs the knowledge state graph based on students' historical learning records and transforms the KT task into a node prediction problem. Given a short-term history exercise record $\mathcal{X}_{\text{Short}}$ and a long-term history exercise record $\mathcal{X}_{\text{Long}}$, the L-SKSKT model processes them from two perspectives (long- and short-term) and constructs two kinds of graphs respectively to capture students' knowledge states better.

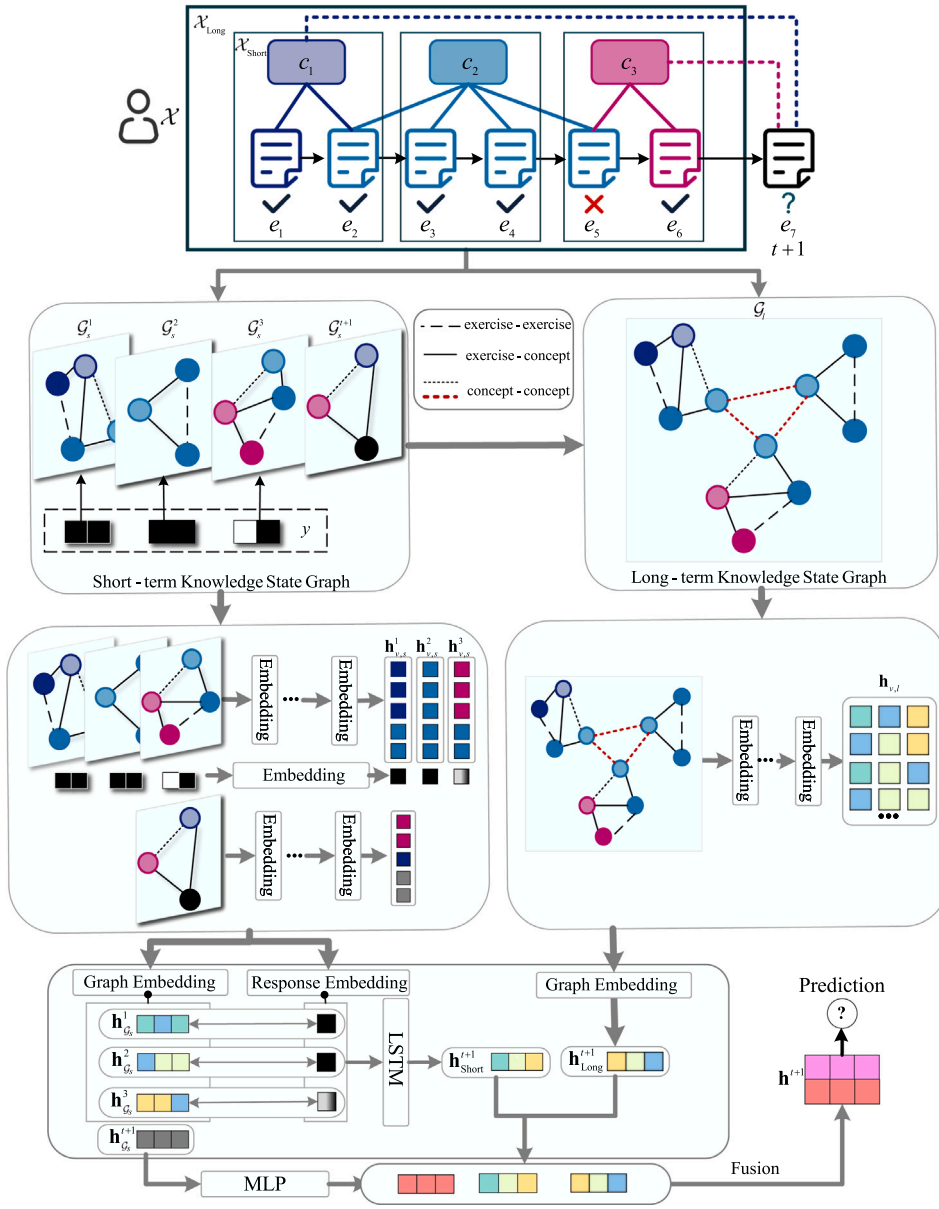


Fig. 2. The overall structure of the proposed L-SKST method.

4.1.1. Short-term knowledge state graph construction

From the short-term perspective, L-SKST mainly focuses on the short dependencies of students' exercise records in a short period of the learning process. Unlike the existing GKT, only considering the *concept-concept* relationship, here we consider more types of relationships such as *exercise-exercise*, *exercise-concept*, *concept-concept*.

Given an arbitrary short-term historical learning record $\mathcal{X}_{\text{Short}}$ and the defined three relationships, We regard all exercises e and concepts c as the node set, and create the relationship for each exercise and the corresponding concept. Specifically, if an exercise e corresponds to a concept c , then create an *exercise-concept* relationship between them (denote as $r_{e,c}$). If e_1 and e_2 contain the same concept c , then create an *exercise-exercise* connection between them (denote as $r_{e,e}$). Likewise, if two concepts c_1 and c_2 share the same exercise e , a *concept-concept* relationship is created between them (denote as $r_{c,c}$). By connecting nodes through three types of relations, the short-term knowledge state graph G_s is constructed. These connections are represented in the adjacency matrix A of it, and since all relationships are undirected, A is symmetric.

4.1.2. Long-term knowledge state graph construction

From the long-term perspective, L-SKST focuses on the long-term dependencies of students' exercise records during a long learning process. Since the length of time M in the $\mathcal{X}_{\text{Long}}$ is usually much larger than the length of time m in the $\mathcal{X}_{\text{Short}}$, so $\mathcal{X}_{\text{Long}}$ contains multiple historical $\mathcal{X}_{\text{Short}}$. In order to capture the long-term dependencies in the long-term learning process, we need to construct a long-term knowledge state graph to associate the historical short-term knowledge state graphs $\mathcal{G}_s$ in it explicitly.

Specifically, for any two $\mathcal{G}_s^i$ and $\mathcal{G}_s^j$, if there exists a common knowledge concept c , we add a temporal connection edge between c^i and c^j (denote as $r_{c,c}^j$). In this way, we can connect $\mathcal{X}_{\text{Short}}$ of different time periods by the common knowledge concepts. As a result, $\mathcal{X}_{\text{Long}}$ can be transformed into a multi-relational graph $\mathcal{G}_l$, namely long-term knowledge state graph, where any $\mathcal{G}_s$ can be regarded as its subgraph, and the same knowledge concepts c in different time periods are regarded as independent nodes. Taking Fig. 2 as an example, there are three short-term knowledge state graphs $\mathcal{G}_s^1$, $\mathcal{G}_s^2$ and $\mathcal{G}_s^3$ for different time periods, with the common knowledge concept c_2 . They are connected by the common knowledge concept c_2 , via the red dashed line in Fig. 2. Although there are large time intervals between $\mathcal{G}_s^1$, $\mathcal{G}_s^2$ and $\mathcal{G}_s^3$, their nodes can still be explicitly connected in $\mathcal{G}_l$. These long-term dependencies at different timestamps cannot be captured in sequence scenarios.

4.2. Knowledge state graph representation learning

To efficiently encode the constructed long- and short-term knowledge state graphs, so we design a knowledge state graph embedding model, which processes the graphs from two perspectives and constructs the corresponding long- and short-term knowledge state representations.

4.2.1. Short-term knowledge state representation

For any $\mathcal{G}_s$, it contains knowledge concepts and exercise nodes that usually have strong dependencies on their neighboring nodes. Therefore, we first consider capturing the short-term dependencies within $\mathcal{G}_s$ and constructing embedding for the exercise concept nodes within it. To be specific, we apply a GCN to aggregate features and generate embeddings for each node in $\mathcal{G}_s$. Formally, the aggregator is defined as follows:

$$\mathbf{h}_{i,s}^{k+1} = \text{ReLU} \left(\frac{1}{|\mathcal{N}_{i,s}^k|} \sum_{j \in \mathcal{N}_{i,s}^k} \mathbf{W}_1^k (\mathbf{h}_{j,s}^k) + \mathbf{W}_2^k \mathbf{h}_{i,s}^k \right) \quad (2)$$

where $\mathcal{N}_{i,s}$ is the set of neighbors of node i in $\mathcal{G}_s$, $\mathbf{W}_1^k$ and $\mathbf{W}_2^k$ are learnable weight parameters used for aggregation and handling the self-loop in the k th layer. After multiple layers of convolution, we can obtain the representation of each node in $\mathcal{G}_s$ which is denoted as $\mathbf{h}_{i,s}$ for the output embedding, by omitting the layer superscripts, $\mathbf{h}_{i,s} \in \mathbb{R}^d$.

For the short-term knowledge state representation, it should reflect the recent knowledge state change. Therefore, we need to combine the node embedding $\mathbf{h}_{i,s}$ and the knowledge mastery of $\mathcal{X}_{\text{Short}}$ to construct the representations that can capture short-term dependencies. Specifically, we first integrate the node embedding $\mathbf{h}_{i,s}$ to construct the graph vector $\mathbf{h}_{\mathcal{G}_s}$ corresponding to $\mathcal{G}_s$, which is constructed as follows:

$$\mathbf{h}_{\mathcal{G}_s} = \text{MEAN} \left(\sum_{i \in \mathcal{G}_s} \mathbf{h}_{i,s} \right) \quad (3)$$

Then, we combine the long short-term memory (LSTM) to construct a short-term knowledge state representation based on the graph-level output $\mathbf{h}_{\mathcal{G}_s}$ with the corresponding $\mathcal{X}_{\text{Short}}$ in which y represents the knowledge mastery level. A student's short-term knowledge state $\mathbf{h}_{\text{Short}}^{t+1}$ is computed based on the representation of the recent timestamp t , which can be obtained in the following way:

$$\mathbf{h}_{\text{Short}}^{t+1} = \text{LSTM} \left(\mathbf{h}_{\text{Short}}^t, \left\{ \left(\mathbf{h}_{\mathcal{G}_s}^1 \parallel \mathbf{h}_{y,s}^1 \right), \left(\mathbf{h}_{\mathcal{G}_s}^2 \parallel \mathbf{h}_{y,s}^2 \right), \dots, \left(\mathbf{h}_{\mathcal{G}_s}^t \parallel \mathbf{h}_{y,s}^t \right) \right\} \right) \quad (4)$$

where $\parallel$ denotes the vector concatenation, $\mathbf{h}_{y,s}$ is the embedding of the response y in the $\mathcal{X}_{\text{Short}}$, $\mathbf{h}_{\mathcal{G}_s}, \mathbf{h}_{y,s} \in \mathbb{R}^d$, and $\mathbf{h}_{\text{Short}}^{t+1} \in \mathbb{R}^{2 \times d}$.

4.2.2. Long-term knowledge state representation

To further capture the implicit long-term dependencies of the exercise concepts across long periods of time, we base on the graph $\mathcal{G}_l$ and execute message propagation and node information integration operations on it. Specifically, the operation measures the interactions between neighboring nodes by introducing an attention mechanism. It is formally expressed as follows:

$$\alpha_{i,j} = \frac{\exp \left(\text{LeakyReLU} \left(\mathbf{a}^T \mathbf{W}_3^k \left[\mathbf{h}_{i,l}^k \parallel \mathbf{h}_{j,l}^k \right] \right) \right)}{\sum_{n \in \mathcal{N}_{i,l}} \exp \left(\text{LeakyReLU} \left(\mathbf{a}^T \mathbf{W}_3^k \left[\mathbf{h}_{i,l}^k \parallel \mathbf{h}_{n,l}^k \right] \right) \right)} \quad (5)$$

where $\mathcal{N}_{i,l}$ denotes the set of neighboring nodes of node i in the $\mathcal{G}_l$, $\mathbf{a}$ and $\mathbf{W}_3^k$ is the trainable weight parameter matrices in k th layer and $\parallel$ is the concatenation operation. Subsequently, we are able to adaptively integrate the embedding of neighboring nodes to generate an embedding of each node in $\mathcal{G}_l$.

$$\mathbf{h}_{i,l}^{k+1} = \text{ReLU} \left(\sum_{j \in \mathcal{N}_{i,l}} \alpha_{i,j} \mathbf{W}_4^k (\mathbf{h}_{j,l}^k) + \mathbf{W}_5^k \mathbf{h}_{i,l}^k \right) \quad (6)$$

where $\mathbf{W}_4$, $\mathbf{W}_5$ are weight parameters to be used for aggregation operations and self-looping operations. After multi-layer operations, we can get the output embedding of each node in $\mathcal{G}_l$, which is denoted as $\mathbf{h}_{i,l}$ for simplicity, $\mathbf{h}_{i,l} \in \mathbb{R}^{2 \times d}$.

Long-term knowledge state representations reflect the system of knowledge that students have accumulated over a long period of time as a result of their learning. Since the long-term dependencies between nodes are captured in $\mathbf{h}_{i,l}$, the embedding is utilized to construct a graph representation under the long-term level, which can be used as a long-term knowledge state representation $\mathbf{h}_{\text{Long}}^{t+1}$ for the students. Formally, $\mathbf{h}_{\text{Long}}^{t+1}$ is constructed as follows:

$$\mathbf{h}_{\text{Long}}^{t+1} = \text{MEAN} \left(\sum_{i \in \mathcal{G}_l} \mathbf{h}_{i,l} \right) \quad (7)$$

4.3. Adaptive fusion and prediction

When students face a new exercise, the long- and short-term knowledge states that they own reinforce each other and together form an integrated knowledge state that provides them with a knowledge base. Therefore, to better predict students' responses on future exercises, it is necessary to adaptively integrate the $\mathbf{h}_{\text{Short}}^{t+1}$ and $\mathbf{h}_{\text{Long}}^{t+1}$ into a unified representation. For this, we use a weighting parameter to fuse these two representations.

$$\mathbf{h}^{t+1} = \rho \odot \mathbf{h}_{\text{Short}}^{t+1} + (1 - \rho) \odot \mathbf{h}_{\text{Long}}^{t+1} \quad (8)$$

where $\odot$ denotes the element multiplication and ρ is the adaptive parameter used to trade-off each student's $\mathbf{h}_{\text{Short}}^{t+1}$ and $\mathbf{h}_{\text{Long}}^{t+1}$.

The goal of KT is regarded as a classification problem, which is to predict the student's future learning performance. Given the unified knowledge state representation $\mathbf{h}^{t+1}$ at $t+1$, the leaning performance y' is calculated as follows:

$$y' = \text{softmax}(\mathbf{W}_6 \mathbf{h}^{t+1} + \mathbf{b}) \quad (9)$$

where $\mathbf{W}_6$, $\mathbf{b}$ are trainable parameters, y' is the category probability. We choose the dimension with the largest value as the prediction category to determine whether the student responses the exercise e in the $\mathcal{X}_{t+1}$ correctly. Following the existing work setting (Cui et al., 2024), we use the cross entropy loss function to minimize the loss between the predicted value y' and the actual value y^{t+1} :

$$\mathcal{L}_1 = \sum - (y^{t+1} \log(y') + (1 - y^{t+1}) \log(1 - y')) \quad (10)$$

Meanwhile, the student's $\mathbf{h}_{\text{Long}}^{t+1}$ should be influenced by the $\mathbf{h}_{\text{Short}}^{t+1}$, and the two representations should exhibit a certain level of consistency. Thus, we introduce a loss $\mathcal{L}_2$ between these two representations, formally:

$$\mathcal{L}_2 = 1 - \frac{\mathbf{h}_{\text{Short}}^{t+1} \cdot \mathbf{h}_{\text{Long}}^{t+1}}{\|\mathbf{h}_{\text{Short}}^{t+1}\| \cdot \|\mathbf{h}_{\text{Long}}^{t+1}\|} \quad (11)$$

For L-SKSKT, our goal is to minimize the total loss $\mathcal{L}$, which consists of the loss $\mathcal{L}_2$ between $\mathbf{h}_{\text{Long}}^{t+1}$ and $\mathbf{h}_{\text{Short}}^{t+1}$ as well as the prediction task loss $\mathcal{L}_1$, combined with the β weight parameter, which is computed as:

$$\mathcal{L} = \mathcal{L}_1 + \beta \mathcal{L}_2 \quad (12)$$

where $\mathcal{L}_1$ guides the optimization of the model parameters by calculating the difference between the predicted label and the true label, with the goal of keeping the predictions as close to the true distribution as possible, and $\mathcal{L}_2$ is constrained by calculating the cosine similarity between the long- and short-term knowledge state representations, thus encouraging consistency between them.

4.4. Complexity analysis

The complexity of our proposed L-SKSKT mainly comes from the following two parts: the short-term knowledge state graph construction and representation, as well as the long-term knowledge state graph construction and representation. For the short-term knowledge state graph construction and representation, the overall complexity is $O((T/m) \cdot k \cdot |\mathcal{G}_s|)$, where $|\mathcal{G}_s|$ denotes the size of each short-term graph, m , T and k represent the short-term window, the length of history exercise records, and the depth of GNN, respectively. While for the long-term knowledge state graph construction and representation, the complexity is $O(k \cdot |\mathcal{G}_l|)$, where $|\mathcal{G}_l|$ is the size of the long-term graph. Therefore, the overall complexity of L-SKSKT is $O((T/m) \cdot k \cdot |\mathcal{G}_s|) + O(k \cdot |\mathcal{G}_l|)$.

In previous graph-based KT models, such as GKT (Nakagawa et al., 2019), the overall complexity is $O(T \cdot k \cdot |\mathcal{G}|)$, where $|\mathcal{G}|$ is the size of the student's knowledge concept graph $\mathcal{G}$. Since GKT updates the representations based on the whole exercise records with length T , so $\mathcal{G}_l \approx \mathcal{G}$. Therefore, the overall complexity of our L-SKSKT is comparable to that of GKT.

5. Experiments

5.1. Datasets and evaluate metrics

We evaluate our method on four public datasets. The statistics for these datasets are shown in Table 3, with detailed explanations as follows:

Table 3
Statistics of all datasets.

Statistics	Datasets			
	ASSIST2009	ASSIST2012	Junyi	Slepemapy
Students	3006	22,397	1000	77,507
Exercise	26,688	53,091	5174	2913
Concepts	107	254	39	1458
Assistment	17,724	46,809	3,827	122,625
Teacher	153	796	228	24
Class	153	1629	41	2
Records	297,343	2,622,857	4,316,340	10,034,618

- **ASSIST2009**¹: ASSIST2009 data is collected based on an online tutoring system that records. We select “problem_id” as “exercise”, “skill_id” as “concepts”, and “teacher”, “class”, and “assessment” as attribute feature data for ‘exercise’. Over the past decade, it has become a widely accepted benchmark for evaluating KT methods.
- **ASSIST2012**²: ASSIST2012 is also collected from the ASSISTments online tutoring system. This is the largest version of the ASSISTments dataset and consists of one year of collected data. We perform the same preprocessing as in ASSIST2009.
- **Junyi**³: Junyi is gathered from the Junyi Research Institute, a Chinese e-learning platform. In line with previous studies, we handpick the top 1000 students with the most extensive Question–Answer records from the log data. Since this dataset does not have feature factors of “assistment”, “class”, and “teacher” in the ASSIST2009, we select 3 feature dimensions of “time_taken”, “count_attempts”, and “point_earned” from the dataset instead. It has 3827 “time_taken”, 228 “count_attempts” and 41 “point_earned”.
- **Slepemapy**⁴: This dataset is collected based on the online system slepemapy.cz. The system uses an adaptive approach to select geo-factual questions. We use “place_asked” as “concepts” and combine it with “type” to construct “exercise”. Also, we use “response_time”, “place_map” and “type” instead of the features it does not have. The dataset has 2 “types”, 122625 “response_times” and 24 “place_maps”.

We evaluate the effectiveness of L-SKSKT using three widely applied evaluation metrics: area under the ROC curve (**AUC**), accuracy (**ACC**), and root mean square error (**RMSE**). AUC measures the model’s ability to distinguish between the samples correctly, ACC assesses the overall prediction accuracy, and RMSE evaluates the variance of predicted values from actual values. Higher AUC and ACC values indicate better performance, while lower RMSE values are superior. For more details regarding the above metrics, please refer to Cui et al. (2024).

5.2. Baselines

Following the literature (Cui et al., 2024; Cui, Yu et al., 2023; Xu, Huang et al., 2023), we compare our method with four groups of KT models: traditional methods, sequence-based modeling methods, attention-based methods, and graph-based methods.

5.2.1. Traditional methods

These methods adopt non-deep learning techniques to predict students’ future performance:

- **BKT** (Corbett, 2000): BKT is grounded in the hidden Markov model and represents students’ knowledge states for each concept as binary variables.
- **IRT** (Rasch, 1981): IRT is a latent variable-based approach that estimates students’ proficiency in specific knowledge or skills by evaluating their responses to test items.
- **KTM** (Sweeney et al., 2016): KTM incorporates the side information about exercises or students into the model by utilizing a factorization machine.

5.2.2. Sequence-based methods

This type of model is based on deep sequence modeling for KT of students’ interaction sequences:

- **DKT** (Piech et al., 2015): The DKT model pioneered the application of deep learning in knowledge tracing. It employs recurrent neural networks to represent students’ knowledge states.
- **AT-DKT** (Liu, Liu et al., 2023): AT-DKT enhances DKT by introducing two auxiliary tasks that predict question tags and student prior knowledge.

¹ <https://sites.google.com/site/assistmentsdata/home/assistment-2009-2010-data/skillbuilder-data-2009-2010>.

² <https://sites.google.com/site/assistmentsdata/home/2012-13-school-data-withaffect>.

³ <https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=1198>

⁴ <https://www.fi.muni.cz/adaptivelearning/data/slepemapy/>.

- **DKT-DSC** (Minn et al., 2018): DKT-DSC extends DKT by clustering student performance using K-means and dynamically updating the classification information as student performance changes.
- **DKVMN** (Zhang, Shi et al., 2017): DKVMN utilizes key memory matrices to store conceptual information and dynamic value memory matrices to track and update the student's proficiency in the corresponding concepts.
- **RCKT** (Cui, Yu et al., 2023): RCKT is based on counterfactual reasoning, on which a novel response influence-based counterfactual KT framework is proposed. RCKT generates response influences by comparing the predicted results of factual sequences with the counterfactual sequences constructed after the intervention.
- **DMKT** (Wang et al., 2021): DMKT estimates students' knowledge acquisition based on their responses and learning materials.
- **LPKT** (Shen et al., 2021): LPKT models the learner's learning process by combining the answer time and inter-response time to calculate the learning gain and forgetting between two consecutive time steps.
- **LBKT** (Xu, Huang et al., 2023): LBKT model explicitly analyzes the effects of learning behaviors, including Speed, Attempts, and Hints, by exploring their impact on students' knowledge states and quantitatively assessing their effectiveness in KT.

5.2.3. Attention-based methods

These types of models are centered on attentional mechanisms for capturing key information from students' historical interactions:

- **AKT** (Ghosh et al., 2020): AKT model uses a monotonic attention mechanism to capture correlations between learning sequences. It uses the Rasch model to combine the individualized difficulty of the problem with the concept.
- **SAKT** (Pandey & Karypis, 2019): SAKT mines feature dependencies implicit in learned features by using a self-attention mechanism.
- **SAINKT** (Choi et al., 2020): SAINT predicts student performance by dividing student interactions into question embedding sequences and response embedding sequences, using an encoder-decoder architecture and scaled dot-product attention mechanism.

5.2.4. Graph-based methods

These methods improve KT by performing graph modeling based on sequences of student interactions:

- **GKT** (Nakagawa et al., 2019): GKT reformulates the KT problem as a time series node-level classification problem and solves it using standard graph learning techniques such as message-passing GNNs.
- **GIKT** (Yang et al., 2021): GIKT effectively fuses exercise-concept associations through GCN and models long-term interactions between students and exercises and concepts.
- **DGEKT** (Cui et al., 2024): DGEKT captures heterogeneous motion-concept associations and interaction transformations through hypergraph and directed graph modeling, and adaptively combines these two graph models using online knowledge distillation techniques, resulting in higher prediction accuracy in KT.

5.3. Implementation details

We use the Pytorch to implement our model and apply the Adam optimizer (Kingma & Ba, 2014) for optimization. Through network training, we conduct a hyperparameter search to determine the optimal configuration. In the experiments, the student's long-term window M is selected from {50, 100, 150, 200}, the student's short-term window m from {10, 20, 30, 40, 50, 80, 100, 150, 200}, the embedding dimension size d from {16, 32, 64, 128, 256}, the number of layers in the knowledge state embedding model from {1, 2, 3}, the learning rate from {0.0005, 0.001, 0.002}, the weight decay coefficient from { $1e-2$, $1e-3$, $1e-4$ }, and the dropout rate from {0.1, 0.2, 0.3}. For each dataset, we divide into a training set and a test set in the ratio of 7:3. Finally, the learning rate is set to 0.001, the weight decay is $1e-3$, the dropout rate is 0.1, the size of short-term window m is 10, the size of the long-term window M is 100, the number of layers of the knowledge state embedding model is 2, the short-term knowledge state embedding dimension is 128, and the long-term knowledge state embedding dimension is 256. For students with sequence lengths less than 100, we populate them with 0. For students with sequence lengths greater than 100, we slice their sequences. Notably, all experiments are performed on Ubuntu 20.04 and A6000*2 GPUs to ensure computational efficiency.

5.4. Experimental results and analysis

Tables 4 and 5 show the results of the experiments on the four versions of ASSIST2009, ASSIST2012, Junyi and Slepemapy under the three evaluation metrics. From them, we can draw the following conclusions:

Graph-based KT methods perform worse than sequence-based KT methods across four datasets. This indicates that previous graph-based KT methods, which focus solely on short-term knowledge states, are insufficient for achieving good predictive performance due to the lack of long-term dependency modeling. Our L-SKSKT achieves better performance than most baseline methods in the three established evaluation metrics (e.g., our model demonstrates the highest AUC performance among the four inductive datasets.).

In terms of AUC, our L-SKSKT outperforms other SOTA models on these four datasets. Compared with the previous SOTA method, RCKT, our L-SKSKT achieves improvements of 5.01%, 3.75%, and 1.34% in AUC on the ASSIST2009, ASSIST2012, and Slepemapy datasets, respectively, along with improvements of 7.1%, 0.89%, and 0.47% in ACC, and reductions of 2.63%, 0.37%, and 0.38% in RMSE. Compared with the SOTA method LBKT, our L-SKSKT achieves improvements of 5.85%, 3.52%, and 0.46% in AUC on the

Table 4

The overall experimental results of baselines vs. our L-SKSKT regarding the evaluation metrics AUC, ACC, and RMSE on ASSIST2009 and ASSIST2012.

Model	ASSIST2009			ASSIST2012		
	AUC	ACC	RMSE	AUC	ACC	RMSE
BKT (Corbett, 2000)	0.6571	0.6209	0.4897	0.6204	0.5983	0.4993
IRT (Rasch, 1981)	0.7510	0.7139	0.4399	0.7431	0.7167	0.4577
KTM (Sweeney et al., 2016)	0.7169	0.6839	0.4335	0.6788	0.6711	0.4774
DKT (Piech et al., 2015) ^a	0.7430	0.7127	0.4372	0.7364	0.7333	0.4226
AT-DKT (Liu, Liu et al., 2023) ^a	0.7574	0.7172	0.4370	0.7544	0.7440	0.4162
DKT-DSC (Minn et al., 2018)	0.7350	0.7131	0.4393	0.7210	0.7007	0.4496
DMKT (Wang et al., 2021) ^a	0.7569	0.7196	0.4370	0.7377	0.7383	0.4224
DKVMN (Zhang, Shi et al., 2017) ^a	0.7400	0.7038	0.4416	0.7351	0.7359	0.4224
LPKT (Shen et al., 2021) ^a	0.7788	0.7235	0.4236	0.7751	0.7567	0.4078
LBKT (Xu, Huang et al., 2023) ^a	0.7863	0.7380	0.4203	0.7823	0.7613	0.4043
RCKT (Cui, Yu et al., 2023)	<u>0.7947</u>	0.7449	<u>0.4109</u>	<u>0.7782</u>	0.7576	0.4101
AKT (Ghosh et al., 2020) ^a	0.7874	<u>0.7689</u>	0.4273	0.7760	0.7559	0.4084
SAKT (Pandey & Karypis, 2019) ^a	0.7111	0.6885	0.4545	0.7335	0.7333	0.4236
SAINKT (Choi et al., 2020)	0.7013	0.6976	0.4511	0.7503	0.7430	0.4116
GKT (Nakagawa et al., 2019)	0.7231	0.7091	0.4575	0.7357	0.7114	0.4211
GIKT (Yang et al., 2021)	0.7896	0.7315	0.4221	0.7754	0.7419	0.4191
DGEKT (Cui et al., 2024)	0.7656	0.7599	0.4198	–	–	–
L-SKSKT	0.8448	0.8159	0.3846	0.8175	0.7665	<u>0.4064</u>

^a Indicates that results are taken from Xu, Huang et al. (2023), all other results are taken from the original paper.

Table 5

The overall experimental results of baselines vs. our L-SKSKT regarding the evaluation metrics AUC, ACC, and RMSE on Junyi and Slepemapy. The IRT, KTM, DKT-DSC, GIKT, and DGEKT models are unable to perform the KT analysis on these two datasets.

Model	Junyi			Slepemapy		
	AUC	ACC	RMSE	AUC	ACC	RMSE
BKT (Corbett, 2000)	0.7100	0.6819	0.4683	0.6920	0.6897	0.4704
DKT (Piech et al., 2015) ^a	0.7586	0.8326	0.3536	0.7813	0.7988	0.4437
AT-DKT (Liu, Liu et al., 2023) ^a	0.7581	0.8325	0.3537	–	–	–
DMKT (Wang et al., 2021) ^a	0.7586	0.8336	0.3538	0.7828	0.8021	0.4209
DKVMN (Zhang, Shi et al., 2017) ^a	0.7565	0.8324	0.3544	0.7799	0.7909	0.4441
LPKT (Shen et al., 2021) ^a	0.7689	0.8344	0.3509	–	–	–
LBKT (Xu, Huang et al., 2023) ^a	<u>0.7723</u>	0.8362	0.3494	–	–	–
RCKT (Cui, Yu et al., 2023)	–	–	–	<u>0.7955</u>	<u>0.8047</u>	<u>0.4054</u>
AKT (Ghosh et al., 2020) ^a	0.7593	0.8325	0.3538	0.7866	0.8019	0.4100
SAKT (Pandey & Karypis, 2019) ^a	0.7590	0.8323	0.3544	0.7850	0.8012	0.4148
GKT (Nakagawa et al., 2019)	0.7589	0.8331	0.3535	0.7837	0.8020	0.4201
L-SKSKT	0.7769	<u>0.8291</u>	<u>0.3499</u>	0.8089	0.8094	0.4016

^a Indicates that results are taken from Xu, Huang et al. (2023), all other results are taken from the original paper.

ASSIST2009, ASSIST2012, and Junyi datasets, respectively. In contrast to previous graph-based KT approaches (e.g., GKT, GIKT and DGEKT), the proposed L-SKSKT can model multiple and complex relationships between exercises and concepts in graphs, modeling the knowledge state from both long- and short-term perspectives. For instance, when compared to DGEKT on the ASSIST2009 dataset, the performance of the three metrics improved by 7.92%, 5.6%, and 3.52%, respectively. And on the ASSIST2012 dataset, there is a performance improvement of 4.21%, 2.46%, and 1.27% for the three metrics compared to GIKT, respectively. The superior performance of L-SKSKT is due to its ability to graphically model students' long- and short-term knowledge states, capture multiple complex relationships between exercise concepts in student's knowledge states to construct representations, and incorporate adaptive fusion mechanisms to improve prediction performance. Experimental results show that our L-SKSKT outperforms current SOTA models.

5.5. Ablation study

To assess the influence of each module within the L-SKSKT model, we carry out an ablation study using four different datasets, and the results can be found in Table 6.

- **w/o Long-term:** In order to validate the importance of capturing the long-term dependencies between exercise concepts for modeling the knowledge state, we only use the $\mathcal{X}_{\text{Short}}$ of the students as a basis for our analysis. Four classes of subvariants are designed for the purpose of the $\mathcal{G}_s$ -construction process: w/o L for keeping the graph construction method unchanged, w/o $L(r_{e,c})$ for deconstructing $r_{e,c}$ during the graph construction process, and w/o $L(r_{c,e})$ as well as w/o $L(r_{e,e})$ for the same reason for deconstructing $r_{c,c}$ and $r_{e,e}$.

Table 6

Ablation study on the four benchmark datasets. Δ denotes the difference between each ablation model and the original L-SKSKT model.

Model	ASSIST2009	ASSIST2012	Junyi	Slepemapy
L-SKSKT	0.8448	0.8175	0.7769	0.8089
w/o Long-term				
w/o L	0.7668	0.7371	0.7021	0.7309
Δ	-0.0780	-0.0804	-0.0748	-0.0780
w/o $L(r_{e,c})$	0.6175	0.5489	0.5117	0.5751
Δ	-0.2274	-0.2686	-0.2652	-0.2338
w/o $L(r_{c,c})$	0.6999	0.6759	0.6557	0.6293
Δ	-0.1449	-0.1416	-0.1212	-0.1796
w/o $L(r_{e,e})$	0.7318	0.6834	0.6791	0.6612
Δ	-0.1130	-0.1341	-0.0978	-0.1477
w/o Short-term				
w/o S	0.7411	0.7001	0.6957	0.6997
Δ	-0.1037	-0.1174	-0.0812	-0.1092
w/o $S(r'_{c,c})$	0.6108	0.5728	0.5990	0.6021
Δ	-0.2340	-0.2447	-0.1779	-0.2068
w/o fusion	0.7595	0.7365	0.7369	0.7889
Δ	-0.0853	-0.0810	-0.0400	-0.0200

- **w/o Short-term:** In this case, the aim is the opposite of the previous one. To confirm the significance of the iteration of students' short-term knowledge states, we only use the $\mathcal{X}_{\text{Long}}$ as the basis for our analysis. We similarly set up several types of subvariants: w/o $S(r'_{c,c})$ which constructs $\mathcal{G}_l$ without constructing $r'_{c,c}$, and for the common concepts merges them into a single node. w/o S represents no changes and only uses the long-term knowledge state as a basis for future predictions.
- **w/o fusion:** Instead of using adaptive fusion, we construct $\mathbf{h}^{t+1}$ directly by concatenating $\mathbf{h}_{\text{Short}}^{t+1}$ and $\mathbf{h}_{\text{Long}}^{t+1}$.

From Table 6, we can see that the performance of all variants of the model decreases to some extent. For all datasets, L-SKSKT outperforms all variants on the evaluation metrics, which fuses long and short-term knowledge states and captures multiple complex relationships of exercise concepts can effectively improve the prediction performance of the KT task. Specifically, the results of w/o L and w/o S show that the model's predictive performance declines to some extent when either the long-term or short-term knowledge states are missing. Additionally, the model's predictive performance is also influenced by the knowledge state graph modeling methods. The various variants indicate that the modeling of the four relationships, $r_{e,c}$, $r_{c,c}$, $r_{e,e}$ and $r'_{c,c}$ significantly affects the model's predictive performance. w/o $L(r_{e,c})$ and w/o $L(r'_{c,c})$ show the greatest decline in performance across the four datasets, reflecting the important roles of $r_{e,c}$ and $r'_{c,c}$ in modeling long- and short-term knowledge states. The results of w/o $L(r_{c,c})$ and w/o $L(r_{e,e})$ also show that incorporating multiple complex relationships of exercise concepts can effectively improve the ability to model the knowledge states. Finally, merely concatenating the long-term and short-term knowledge state representations in a linear fashion may not produce optimal results; L-SKSKT achieves better performance than w/o fusion by using an adaptive fusion module.

5.6. Parameter sensitivity analysis

To study the sensitivity of L-SKSKT, we evaluate the impact of key hyperparameters on its performance. Specifically, we consider the size of the long-term window M and the short-term window m , the graph embedding size d and the loss weight β .

We conduct experiments on four datasets for different parameter combinations of M and m , and the results are shown in Fig. 3. From the results in Fig. 3, we have the following conclusions: (1) Increasing the short-term window m leads to a significant decline in model performance. For all datasets, when M is fixed, as m increases, the model's performance gradually decreases. When $m = M$, the model performance decreases significantly. The reason may be that setting the short-term window m too big makes L-SKSKT unable to effectively distinguish the long- and short-term knowledge states as well as increases the computational cost. (2) An appropriate long-term window size of M can effectively enhance the L-SKSKT to model long-term knowledge states. The experimental results show that the size of the long-term window M has a significant impact on model performance. Both too big and too small values of M lead to performance degradation. Based on cognitive theory (Krathwohl, 2002; Paas et al., 2016) and extensive empirical analysis (Cui, Yu et al., 2023; Xu, Huang et al., 2023), we find that students' learning preferences are dynamic. Generally, the long-term window size required for students to form a stable long-term knowledge state is around 100. (3) The optimal balance between M and m can effectively improve model performance and can also ensure that L-SKSKT captures the dynamic iterative features of short-term knowledge states as well as accurately models long-term dependencies in the exercise records. (4) The performance of the model on different datasets shows different levels of sensitivity to changes in M and m . Compared to the ASSIST2009 and Junyi datasets, the ASSIST2012 and Slepemapy datasets are more sensitive to changes in m and M , with a significant decrease in AUC values. This suggests that L-SKSKT experiences slight differences in the impact of the sizes of M and m on the modeling of the long- and short-term knowledge states across different datasets. Overall, the model performs best when $M = 100$ and $m = 10$. Regarding the embedding dimension d and loss weight β , we verified the sensitivity of these two parameters on the ASSIST2009 dataset, and the results are shown in Fig. 4. The results show that the best results are obtained when the dimension is 128 and the loss weight is 0.1. These two settings are relatively common values.

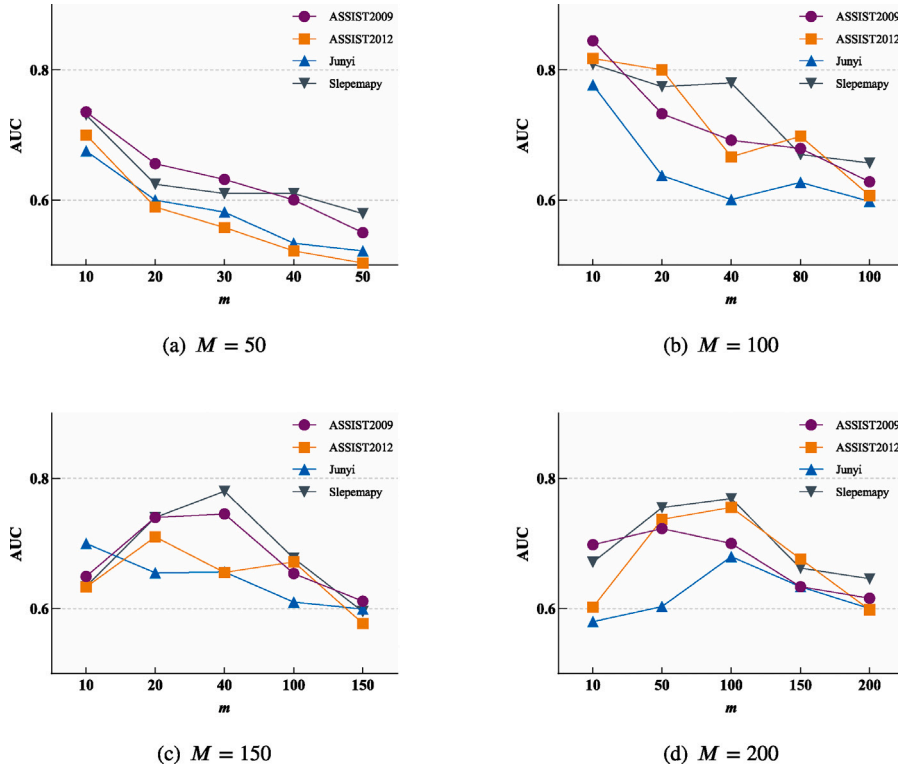


Fig. 3. Subfigures (a)–(d) shows different combinations of M and m vs. AUC values under the four datasets.

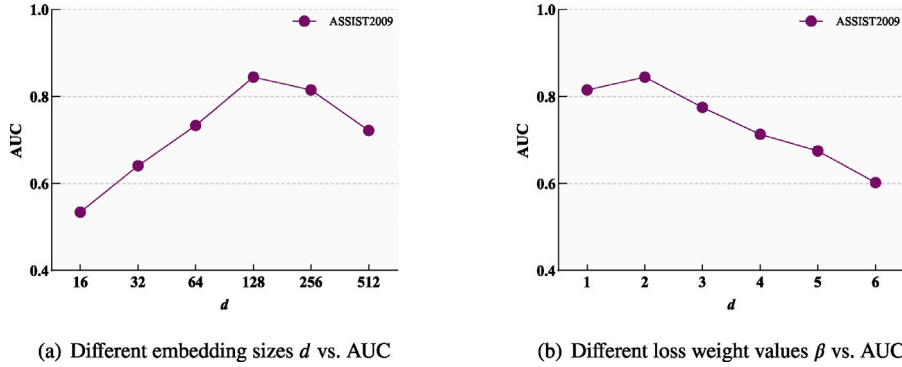


Fig. 4. Subfigures (a)–(b) denote different embedding sizes d and loss weight values β vs. AUC values under the ASSIST2009 dataset.

5.7. Visualization of KT prediction performance

To better represent the advantages of L-SKSKT in modeling student's long- and short-term knowledge states. We extracted sequences of student-specific exercise records of length 100 from the ASSISTment2009 dataset. By dividing them into subsequence data of length $\{\mathcal{X}_{10}, \mathcal{X}_{20}, \mathcal{X}_{30}, \dots, \mathcal{X}_{90}, \mathcal{X}_{100}\}$, the predictions of responses to future exercises are performed on them separately. We set up different comparisons: the GKT model based on knowledge-concept relations, the GIKT model based on long-term interactions, and the DGEKT model based on heterogeneous motion associations and interaction transitions. Finally, all results are shown in Fig. 5.

From this heat map, it can be observed that the predicted probabilities of L-SKSKT are highly consistent with students' true response labels, showing its superior performance in predicting students' responses to the exercise. Compared with GKT, L-SKSKT shows better results on all probability metrics, which suggests that L-SKSKT captures the multiple complex relationships of exercise concepts by modeling long- and short-term knowledge states and effectively avoids the introduction of irrelevant knowledge features, thus improving the accuracy of KT analysis. In addition, L-SKSKT outperforms GIKT in terms of coverage of student knowledge state

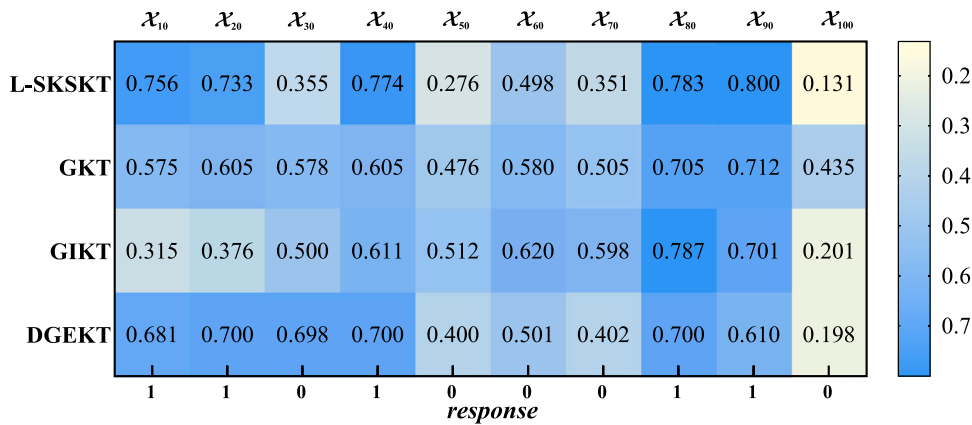


Fig. 5. A visual tracing example of a student from ASSSISTment2009. In this heatmap, the y-axis represents the different models, the x-axis represents the student's real response, and the values in the graph represent the probability that the student gives the correct response.

representations. GIKT performs poorly when learning sequences are short, and thus relying on long-term knowledge alone may lead to performance deficits when exercise records are limited. Compared with DGEKT, our model is roughly equal in performance, but on shorter sequence data, L-SKSKT has a clear advantage in knowledge state modeling with lower error rates in predicting probabilities. In summary, L-SKSKT provides a more robust method for graph-based KT studies.

6. Conclusions and future works

In this paper, we combine the cognitive load theory to analyze the knowledge state from both long- and short-term perspectives and explore its impact on KT. To this end, we introduce a knowledge tracing model based on long and short-term knowledge state representations (L-SKSKT). Specifically, we first model student's knowledge states from long- and short-term perspectives, and construct the corresponding knowledge state graphs based on the multiple complex relationships among the exercise concepts therein. Subsequently, we construct a knowledge state graph embedding model to quantify the long- and short-term dependencies between the exercise concepts and generate the corresponding long- and short-term knowledge state representations. Finally, we introduce an adaptive fusion module that dynamically adjusts the fusion weights of these two knowledge state representations to construct a unified representation to characterize students' real knowledge states. Our experiments on public datasets show that compared with the current SOTA sequence KT model, the AUC is improved by 5.85%, and compared with the SOTA graph-based KT model, the AUC is improved by 7.92%, demonstrating the effectiveness and efficiency of our proposed model.

There are four ways in which this work could be further continued. First, we plan to develop a more adaptive and complex graph construction scheme, particularly one that can incorporate temporal variation features. Second, during the modeling process, we will conduct a detailed analysis of how the sequence in which students master knowledge concepts affects their performance. By studying the order of knowledge concept acquisition, we can more accurately predict students' future performance, thereby optimizing the KT model. Additionally, the emergence of pre-trained models and large language models (LLMs) provides new insights for KT research. We will explore these network architectures in depth to design more effective embedding methods that enhance the model's ability to accurately capture and predict students' knowledge states. Afterward, we will introduce more sophisticated forgetting mechanisms and model the pattern of knowledge forgetting during the learning process, thereby enhancing our graph representation building block to provide more accurate knowledge state assessment and prediction. Finally, to improve the interpretability and the generalization of the model, we plan to develop a visualization and analysis tool that can parse the model modeling process. It can reveal the role of key nodes and edges in the student's knowledge state graph and explain the key factors behind the predicted results. Meanwhile, in our future work, we make the experimental goal of the model is not limited to the communal dataset, but also to be applied on the dataset of mathematics, art, biology, and so on, to enhance the generalization of the model. By improving the interpretability and the generalization of the model, we hope to be able to support personalized teaching decisions more effectively and provide more practical value to the educational practice.

CRedit authorship contribution statement

Ganfeng Yu: Writing – review & editing, Writing – original draft, Validation, Software, Methodology. **Zhiwen Xie:** Writing – review & editing, Methodology. **Guangyou Zhou:** Writing – review & editing, Supervision, Investigation, Funding acquisition. **Zhuo Zhao:** Writing – review & editing. **Jimmy Xiangji Huang:** Writing – review & editing.

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Data availability

Data will be made available on request.

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